

Update

- We have considered
  - Angle modulation
- We now consider
  - **Noise in communication systems**

Noise

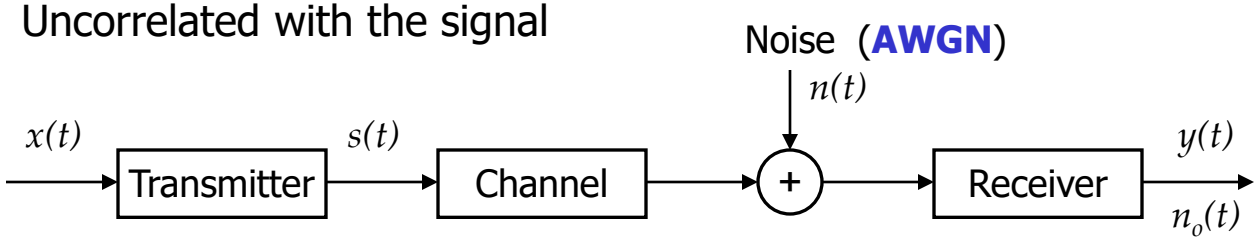
Noise : *random fluctuations* of power in systems

→ Corrupt the signal waveform

→ Degrade the performance of communication systems

Sources of Noise: random wandering of free electrons in resistors (thermal noise), random flow of charges in semiconductor junction (shot noise), etc.

- **A**dditive noise
- Zero-mean **W**hite **G**aussian-distributed **N**oise, power spectral density $S_n(f)=\eta/2$
- Uncorrelated with the signal

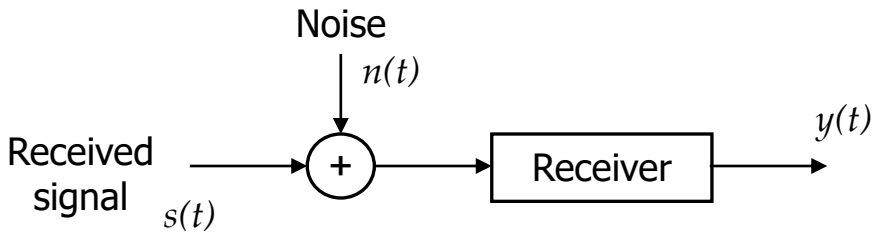


Performance measures: Analog → ***Signal-to-Noise Ratio*** (**SNR**)

Digital → ***Bit-Error-Rate*** (**BER**)

Noises: Signal-to-Noise Ratio

Assume the channel introduces no distortion and receiver is linear



$$SNR_o = \frac{\text{average power of message signal the receiver output}}{\text{average power of noise at the receiver output}}$$

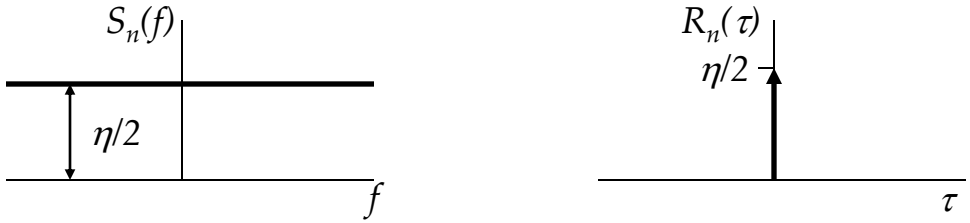
$$y(t) = s_o(t) + n_o(t)$$

where $s_o(t)$ and $n_o(t)$ are the signal and noise waveforms at the receiver output.

$$SNR_o = \frac{\overline{s_o^2}}{\overline{n_o^2}}$$

White Noise and Filtered Noise

White noise: flat spectral density over a wide range of frequencies

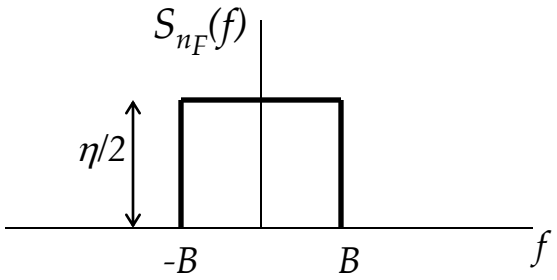


Filtered white noise:

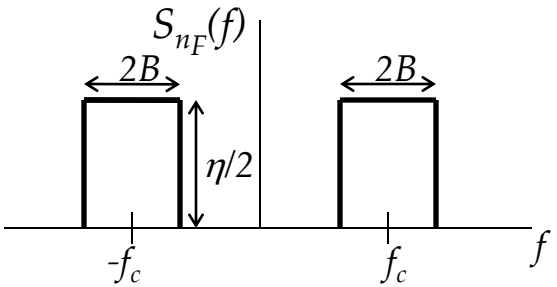


$$S_{n_F}(f) = |H(f)|^2 S_n(f) = \frac{\eta}{2}|H(f)|^2$$

$$\overline{n_F^2} = \int_{-\infty}^{\infty} S_{n_F}(f)df = \frac{\eta}{2}\int_{-\infty}^{\infty}|H(f)|^2 df$$



Low-pass filter

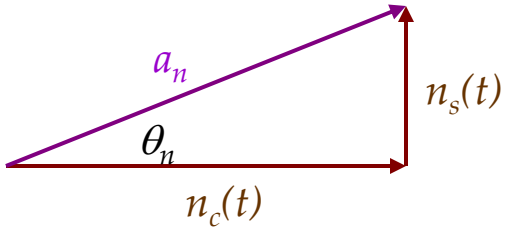


band-pass filter

Bandpass Noise

For bandpass systems, which limit the bandwidth of the noise, some of the noise fluctuations are restricted.

As the bandwidth of the noise becomes small compared to the center frequency, it becomes possible to approximate it with a phasor representation.



$$\theta_n = \tan^{-1}\left(\frac{n_s(t)}{n_c(t)}\right)$$

Here $n_c(t)$ is that portion of the noise which is the in-phase component and $n_s(t)$ is the quadrature component in the phasor representation.

The complete phasor representation of the **narrowband random noise** is

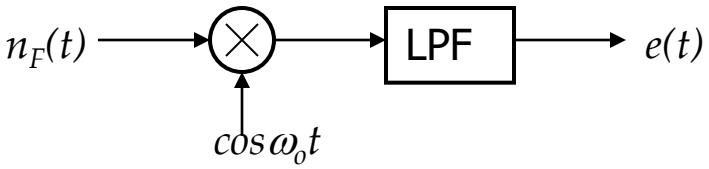
$$[n_c(t) + jn_s(t)]e^{j\omega_o t}$$

where ω_o is the center frequency.

$$n_F(t) = \mathrm{Re}\{[n_c(t) + jn_s(t)]e^{j\omega_o t}\} = n_c(t)\cos\omega_o t - n_s(t)\sin\omega_o t$$

which is known as the bandpass representation of noise.

Bandpass Noise



$$n_F(t)\cos\omega_o t = n_c(t)\cos^2\omega_0 t - n_s(t)\sin\omega_0 t\cos\omega_0 t$$
$$= \tfrac{1}{2}n_c(t)[1+\cos 2\omega_0 t] - \tfrac{1}{2}n_s(t)\sin 2\omega_0 t$$

The output of the low-pass filter is

$$e(t) = [n_F(t)\cos\omega_0 t]_{LP} = \frac{1}{2}n_c(t)$$

The power spectral density of n_c is

$$S_{n_c}(\omega) = \lim_{T\to\infty}\frac{\left|\Im\{n_{cT}(t)\}\right|^2}{T} = \lim_{T\to\infty}\frac{\left|\Im\{2[n_{FT}(t)\cos\omega_0 t]_{LP}\}\right|^2}{T}$$

$$S_{n_c}(\omega) = \left\{\lim_{T\to\infty}\left[\frac{\left|N_{FT}(\omega-\omega_0)+N_{FT}(\omega+\omega_0)\right|^2}{T}\right]\right\}_{LP}$$

Bandpass Noise

For random noise, the average of the cross products goes to zero, thus

$$S_{n_c}(\omega) = [S_{n_F}(\omega - \omega_0) + S_{n_F}(\omega + \omega_0)]_{LP}$$

The case for n_s follows in the same manner by considering $n_F(t) sin\ \omega_c t$, yielding

$$S_{n_s}(\omega) = [S_{n_F}(\omega - \omega_0) + S_{n_F}(\omega + \omega_0)]_{LP}$$

Hence, $S_{n_c}(\omega) = S_{n_s}(\omega) = [S_{n_F}(\omega - \omega_0) + S_{n_F}(\omega + \omega_0)]_{LP}$

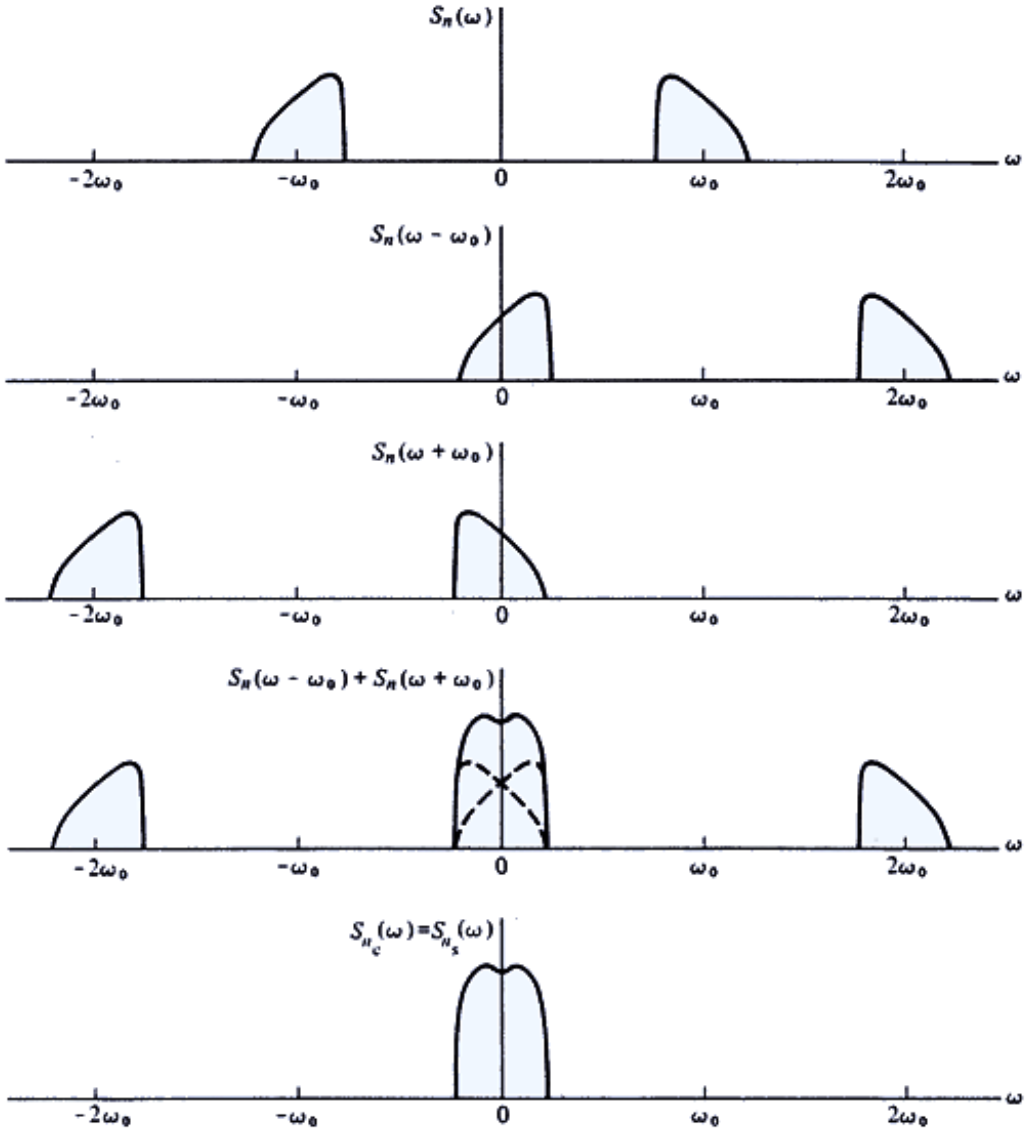
Thus, their mean-square values are equal;

$$\overline{n_F^2(t)} = \overline{n_c{}^2(t)} = \overline{n_s{}^2(t)}$$

The mean-square value of bandpass random noise is

$$\overline{n_F^2(t)} = \tfrac{1}{2}\overline{n_c{}^2(t)} + \tfrac{1}{2}\overline{n_s{}^2(t)} \qquad \because n_F(t) = n_c(t)\cos\omega_o t - n_s(t)\sin\omega_o t$$

Bandpass Noise



Bandpass Noise

Example: White noise with power spectral density $\eta/2$ is filtered by a rectangular bandpass filter with $H(f)=1$, centered at f_o and having a bandwidth B. Find the power spectral density of $n_c(t)$ and $n_s(t)$. Calculate the power in $n_c(t)$, $n_s(t)$ and $n_F(t)$.

Solution: Since the filter is rectangular with $H(f)=1$, the power spectral density of the output noise $n(t)$ is:

$$S_{n_F}(f)=\begin{cases}\eta/2 & f_o-B/2\le|f|\le f_o+B/2\\ 0 & elsewhere\end{cases}$$

Hence, $S_{n_F}(f+f_o)=S_{n_F}(f-f_o)$ and the spectral densities of $n_c(t)$ and $n_s(t)$ are

$$S_{n_c}(f)=S_{n_s}(f)=S_{n_F}(f-f_o)+S_{n_F}(f+f_o)=\eta/2+\eta/2=\eta \qquad \text{for } |f|\le B/2$$

Powers of $n_c(t)$ and $n_s(t)$ are

$$\overline{n_c^2}=\overline{n_s^2}=\int_{-B/2}^{B/2}S_{n_c}(f)df=\eta B$$

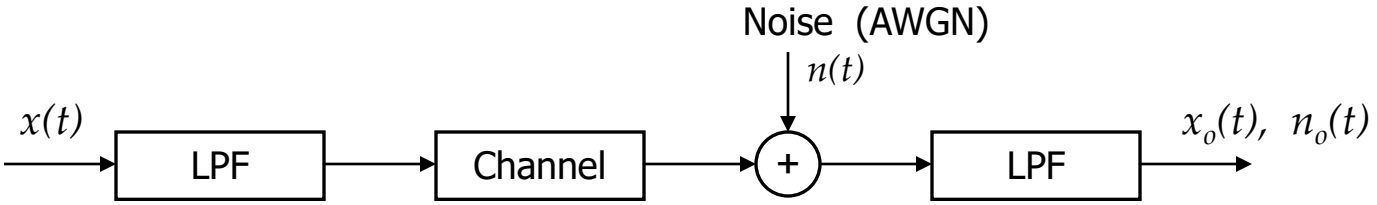
Power of $n_F(t)$ is

$$\overline{n_F^2}=\int_{-f_o-B/2}^{-f_o+B/2}S_n(f)df+\int_{f_o-B/2}^{f_o+B/2}S_n(f)df=2\left(\frac{\eta}{2}\right)B=\eta B$$

Noise in Baseband Comm. Systems

Baseband transmission: signal is transmitted without any modulation.

Assume both the transmitter filter and receiver filter are ideal low-pass filters with bandwidth B



Assume distortionless channel: $x_o(t) = Kx(t - t_d)$

where K and t_d are the amplification/attenuation and time-delay, respectively.

Average output signal power: $\overline{x_o^2} = P_R$ (received power)

Average output noise power: $\overline{n_o^2} = \int_{-B}^{B} S_n(f)df = \int_{-B}^{B} \left(\frac{\eta}{2}\right) df = \eta B$

$$SNR_o = \frac{P_R}{\eta B}$$

Noise in Baseband Comm. Systems

Example: Consider an analog baseband communication system with additive white noise. The transmission channel is assumed to be distortionless and the power spectral density of white noise ($\eta/2$) is 10^{-9} watt per hertz (W/Hz). The signal to be transmitted is an audio signal with 4 kHz bandwidth. At the receiver end, an RC low pass filter with a 3-dB bandwidth of 8 kHz is used to limit the noise power at the output. **Calculate the output noise power**.

Solution: For a RC low-pass filter (3-dB bandwidth=8kHz), the frequency response is

$$H(\omega)=\frac{1/(j\omega C)}{R+1/(j\omega C)}=\frac{1}{1+j\,\omega/\omega_o} \quad where \quad \omega_o=1/(RC)$$

$$|H(\omega)|=\frac{1}{\left[1+(\omega/\omega_o)^2\right]^{1/2}}$$

3dB bandwidth at $\omega=\omega_o$ → = $\omega_o=2\pi(8000)$

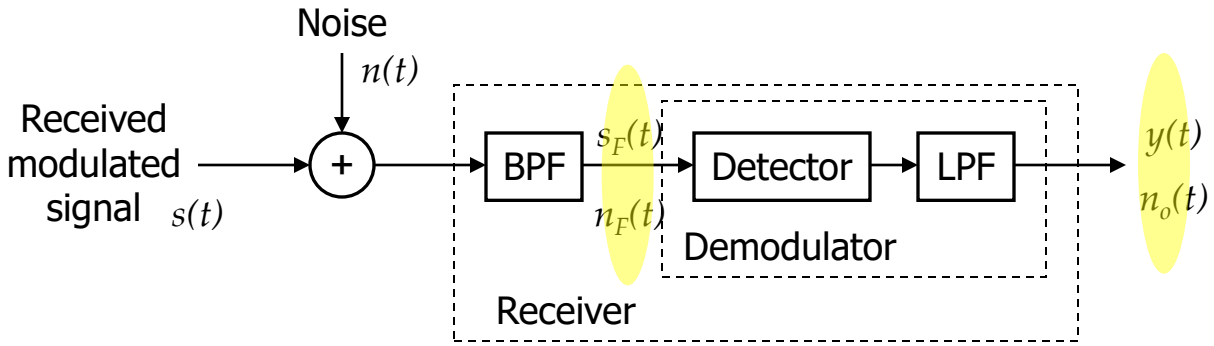
Average output noise power:

$$\overline{n_o^2}=\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{\eta}{2}\right)|H(\omega)|^2\,d\omega=\frac{\eta}{2}\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{1+(\omega/\omega_o)^2}\,d\omega$$

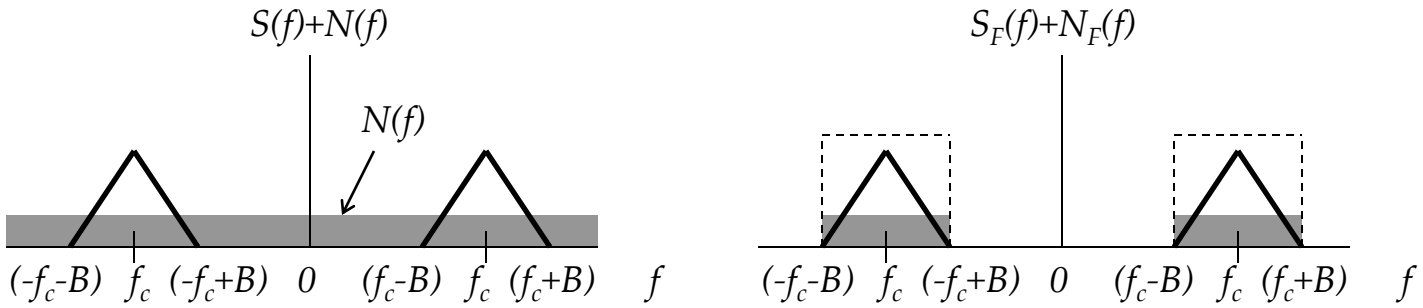
$$\overline{n_o^2}=\frac{\eta}{2}\frac{1}{2\pi}(\pi\omega_o)=\frac{1}{4}\eta\omega_o$$

$$\overline{n_o^2}=\frac{1}{4}(2(10^{-9})(2\pi)(8\times10^3)=25.2\mu W$$

Noise in Amplitude Modulated Systems



The bandpass filter limits the amount of noise outside signal band that reaches the demodulator ("out-of-band" noise)



Narrowband filtered noise: $n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$

Noise power: $\overline{n_c^2} = \overline{n_s^2} = \overline{n_F^2} = 2\eta B$

Received signal power: $P_R = \overline{s_F^2(t)}$

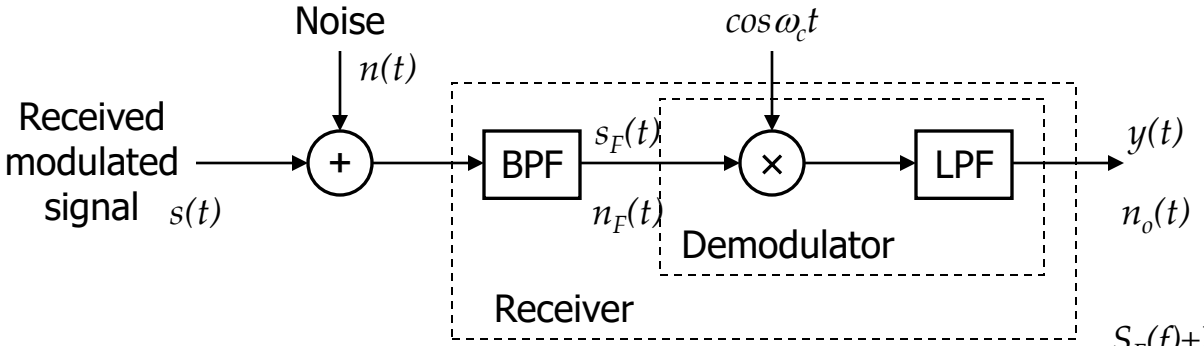
Noise in DSB-SC Systems

In a DSB-SC system, the received modulated signal is:

$$s(t) = A_c x(t) \cos \omega_c t$$

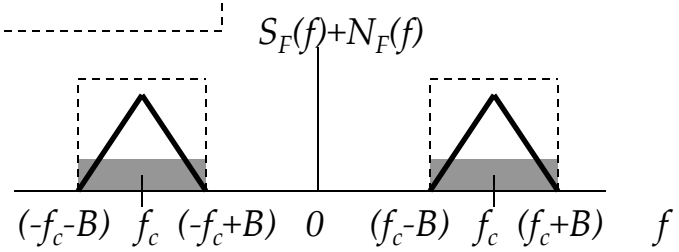
where $x(t)$ is the message signal which is band-limited with bandwidth B

Consider the demodulation by *synchronous detection*:



$$s_F(t) = A_c x(t) \cos \omega_c t$$

$$n_F(t) = n_c(t) \cos \omega_c t - n_s(t) \sin \omega_c t$$



Input signal power to the demodulator: $\overline{s_F^2(t)} = A_c^2 \overline{x^2(t) \cos^2 \omega_c t} = \frac{1}{2} A_c^2 \overline{x^2(t)} = P_R$

Input noise power to the demodulator: $\overline{n_F^2(t)} = 2\eta B$

Noise in DSB-SC Systems

Output signal after LPF: $y(t) = [s_F(t)\cos\omega_c t]_{LP} = \frac{A_c}{2}x(t)$

Output signal power after LPF: $\overline{y^2(t)} = \frac{A_c^2}{4}\overline{x^2(t)}$

Output noise after LPF:

$$n_o(t) = [n_F(t)\cos\omega_c t]_{LP}$$
$$= [n_c(t)\cos^2\omega_c t - n_s(t)\sin\omega_c t\cos\omega_c t]_{LP}$$
$$= \{\tfrac{1}{2}n_c(t)[1+\cos 2\omega_c t] - \tfrac{1}{2}n_s(t)\sin 2\omega_c t\}_{LP}$$
$$= \tfrac{1}{2}n_c(t)$$

Output noise power after LPF: $\overline{n_o^2(t)} = \frac{1}{4}\overline{n_c^2(t)} = \frac{1}{4}\overline{n_F^2(t)}$ $\quad \because \overline{n_c^2} = \overline{n_s^2} = \overline{n_F^2}$

SNR before demodulator: $SNR_i = \frac{\overline{s_F^2}}{\overline{n_F^2}} = \frac{\frac{1}{2}A_c^2\overline{x^2(t)}}{\overline{n_F^2}} = \frac{P_R}{\overline{n_F^2}} = \frac{P_R}{2\eta B}$

Output SNR after demodulator: $SNR_o = \frac{\overline{y^2(t)}}{\overline{n_o^2(t)}} = \frac{\frac{1}{4}A_c^2\overline{x^2(t)}}{\frac{1}{4}\overline{n_F^2(t)}} = \frac{2P_R}{2\eta B} = 2SNR_i$

The detector (demodulator) improves the SNR in a DSB-SC system by **a factor of 2**. This improvement results from the fact that the **coherent detector rejects the quadrature noise components** in the input noise, thereby halving the mean-square noise power.

$$SNR_o = \frac{P_R}{\eta B}$$ **for DSB-SC**

Noise in DSB-SC Systems

Example: A DSB-SC system with additive white noise is demodulated by a synchronous detector with a phase error ϕ, i.e. $cos(\omega_c t + \phi)$. The original message signal is $x(t)$. **Show that** $SNR_o = \gamma \cos^2 \phi$ where γ is the output SNR when the synchronous detector has no phase error.

$$s_F(t) = A_c x(t) \cos \omega_c t$$

$$n_F(t) = n_c(t) \cos \omega_c t - n_s(t) \sin \omega_c t$$

$$\gamma = \frac{A_c^2 \overline{x^2(t)}}{\overline{n_F^2(t)}}$$

Solution:

Output signal after LPF: $y(t) = \left[s_F(t) \cos(\omega_c t + \phi) \right]_{LP} = \frac{A_c}{2} x(t) \cos \phi$

Output signal power after LPF: $\overline{y^2(t)} = \frac{A_c^2}{4} \overline{x^2(t)} \cos^2 \phi$

Output noise after LPF:

$$n_o(t) = \left[n_F(t) \cos(\omega_c t + \phi) \right]_{LP}$$
$$= \left[\tfrac{1}{2} n_c(t) \cos \phi + \tfrac{1}{2} n_s(t) \sin \phi \right]$$

Output noise power after LPF:

$$\overline{n_o^2(t)} = \tfrac{1}{4} \overline{n_c^2(t)} \cos^2 \phi + \tfrac{1}{4} \overline{n_s^2(t)} \sin^2 \phi$$
$$= \tfrac{1}{4} \overline{n_F^2(t)} \left(\cos^2 \phi + \sin^2 \phi \right)$$
$$= \tfrac{1}{4} \overline{n_F^2(t)}$$

$$\because \overline{n_c^2} = \overline{n_s^2} = \overline{n_F^2}$$

Output SNR after demodulator:

$$SNR_o = \frac{\overline{y^2(t)}}{\overline{n_o^2(t)}} = \frac{\tfrac{1}{4} A_c^2 \overline{x^2(t)} \cos^2 \phi}{\tfrac{1}{4} \overline{n_F^2(t)}} = \frac{A_c^2 \overline{x^2(t)}}{\overline{n_F^2(t)}} \cos^2 \phi = \gamma \cos^2 \phi$$

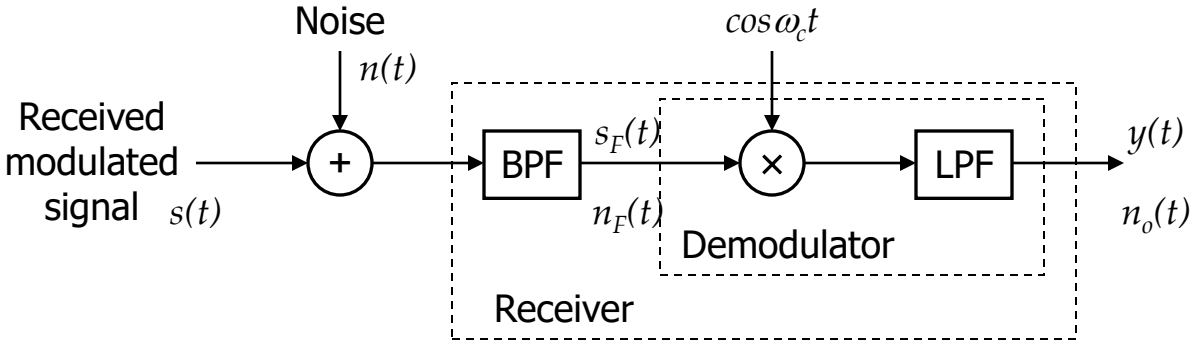
Noise in SSB Systems

In SSB system, the received modulated signal is:

$$s(t) = A_c\left[x(t)\cos\omega_c t + \hat{x}(t)\sin\omega_c t\right]$$

where $x(t)$ is the message signal which is band-limited with bandwidth B

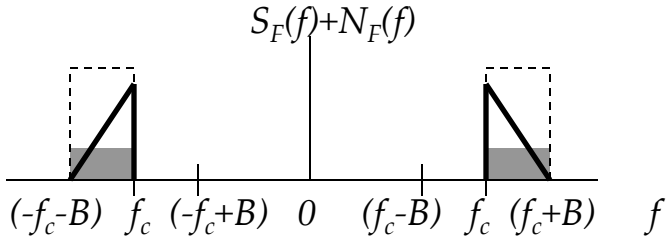
Consider the demodulation by *synchronous detection*:



$$s_F(t) = A_c\left[x(t)\cos\omega_c t + \hat{x}(t)\sin\omega_c t\right]$$

$$n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$$

$$\overline{n_c^2} = \overline{n_s^2} = \overline{n_F^2} = \eta B$$



Noise in SSB Systems

Input signal to the demodulator: $s_F(t) = A_c\left[x(t)\cos\omega_c t + \hat{x}(t)\sin\omega_c t\right]$

Input noise to the demodulator: $n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$

Input signal power to the demodulator:

$$\begin{aligned}\overline{s_F^2(t)} &= A_c^2\left[\overline{x^2(t)\cos^2\omega_c t + \hat{x}^2(t)\sin^2\omega_c t + 2x(t)\hat{x}(t)\sin\omega_c t\cos\omega_c t}\right] \\ &= A_c^2\left[\overline{x^2(t)}\,\overline{\cos^2\omega_c t} + \overline{\hat{x}^2(t)}\,\overline{\sin^2\omega_c t}\right] \\ &= A_c^2\left[\overline{x^2(t)}\,\tfrac{1}{2} + \overline{x^2(t)}\,\tfrac{1}{2}\right] \\ &= A_c^2\,\overline{x^2(t)} = P_R\end{aligned}$$

Input noise power to the demodulator: $\overline{n_F^2(t)} = \eta B$

Output signal to the demodulator: $y(t) = \left[s_F(t)\cos\omega_c t\right]_{LP} = \dfrac{A_c}{2}x(t)$

Output noise to the demodulator: $n_o(t) = \left[n_F(t)\cos\omega_c t\right]_{LP} = \tfrac{1}{2}n_c(t)$

Output signal power to the demodulator: $\overline{y^2(t)} = \dfrac{A_c^2}{4}\overline{x^2(t)}$

Output noise power to the demodulator: $\overline{n_o^2(t)} = \tfrac{1}{4}\overline{n_c^2(t)} = \tfrac{1}{4}\overline{n_F^2(t)}$

Noise in SSB Systems

SNR before demodulator: $SNR_i = \dfrac{\overline{s_F^2}}{\overline{n_F^2}} = \dfrac{A_c^2 \overline{x^2(t)}}{\overline{n_F^2}} = \dfrac{P_R}{\eta B}$

Output SNR after demodulator: $SNR_o = \dfrac{\overline{y^2(t)}}{\overline{n_o^2(t)}} = \dfrac{\frac{1}{4} A_c^2 \overline{x^2(t)}}{\frac{1}{4} \overline{n_F^2(t)}} = \dfrac{P_R}{\eta B} = SNR_i$

The detector (demodulator) does not improve the SNR in a SSB-SC system.

$$SNR_o = \frac{P_R}{\eta B} \qquad \textbf{for SSB}$$

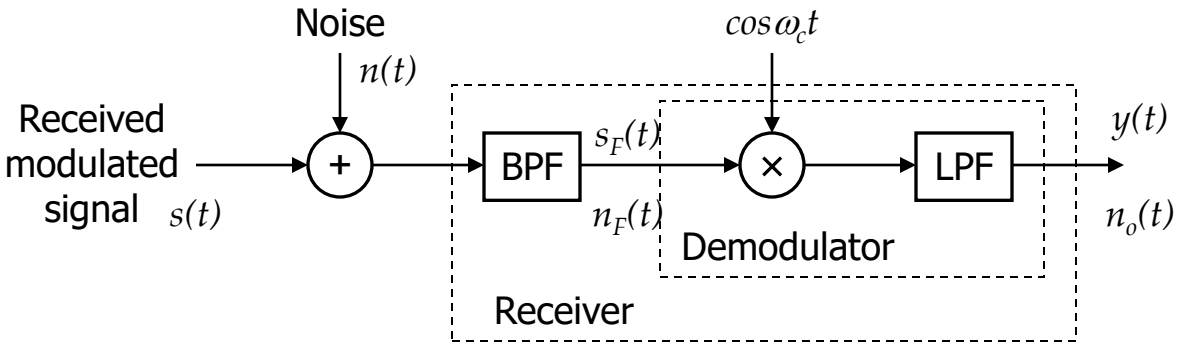
Noise in DSB-LC (AM) Systems

In DSB-LC (AM) system, the received modulated signal is:

$$s(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t$$

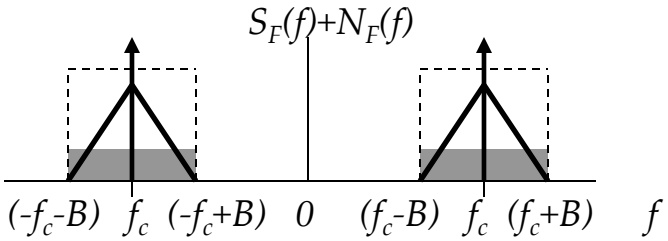
where $x(t)$ is the message signal which is band-limited with bandwidth B

Consider the demodulation by ***synchronous detection***:



$$s_F(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t$$

$$n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$$



Noise in DSB-LC (AM) Systems

Input signal to the demodulator: $s_F(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t$

Input noise to the demodulator: $n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$

Input signal power to the demodulator:

$$\begin{aligned}\overline{s_F^2(t)} &= A_c^2\overline{\left[1 + m_a x(t)\right]^2}\,\overline{\cos^2\omega_c t} \\ &= \tfrac{1}{2}A_c^2\left[1 + m_a^2\overline{x^2(t)} + 2m_a\overline{x(t)}\right] \\ &= \tfrac{1}{2}A_c^2\left[1 + m_a^2\overline{x^2(t)}\right] \quad \text{assume } \overline{x(t)} = 0 \quad (\text{zero mean message signal}) \\ &= P_R\end{aligned}$$

Input noise power to the demodulator: $\overline{n_F^2(t)} = 2\eta B$

Output signal to the demodulator: $y(t) = \left[s_F(t)\cos\omega_c t\right]_{LP} = \tfrac{1}{2}A_c\left[1 + m_a x(t)\right]$

By removing the DC term, gives $y(t) = \tfrac{1}{2}A_c m_a x(t)$

Output noise to the demodulator: $n_o(t) = \left[n_F(t)\cos\omega_c t\right]_{LP} = \tfrac{1}{2}n_c(t)$

Noise in DSB-LC (AM) Systems

Output signal power to the demodulator: $\overline{y^2(t)} = \frac{A_c^2}{4} m_a^2 \overline{x^2(t)}$

Output noise power to the demodulator: $\overline{n_o^2(t)} = \frac{1}{4}\overline{n_c^2(t)} = \frac{1}{4}\overline{n_F^2(t)}$

SNR before demodulator: $SNR_i = \frac{\overline{s_F^2}}{\overline{n_F^2}} = \frac{P_R}{2\eta B}$

Output SNR after demodulator:

$$SNR_o = \frac{\overline{y^2(t)}}{\overline{n_o^2(t)}} = \frac{\frac{1}{4}A_c^2 m_a^2 \overline{x^2(t)}}{\frac{1}{4}\overline{n_F^2(t)}} \cdot \frac{1 + m_a^2 \overline{x^2(t)}}{1 + m_a^2 \overline{x^2(t)}}$$

$$= \frac{\frac{1}{2}m_a^2 \overline{x^2(t)}}{\frac{1}{4}(2\eta B)} \cdot \frac{\frac{1}{2}A_c^2\left[1 + m_a^2 \overline{x^2(t)}\right]}{1 + m_a^2 \overline{x^2(t)}}$$

$$= \frac{m_a^2 \overline{x^2(t)}}{1 + m_a^2 \overline{x^2(t)}} \cdot \frac{P_R}{\eta B}$$

$$SNR_o = \frac{m_a^2 \overline{x^2(t)}}{1 + m_a^2 \overline{x^2(t)}} \cdot \frac{P_R}{\eta B}$$ **For DSB-LC**

As $m_a^2 \overline{x^2(t)} \le 1$ to avoid distortion (over-modulation)

$$\Rightarrow \quad SNR_o \le \frac{1}{2}\frac{P_R}{\eta B}$$

SNR_o of DSB-LC is at least 3-dB worse than that in DSB-SC and SSB systems

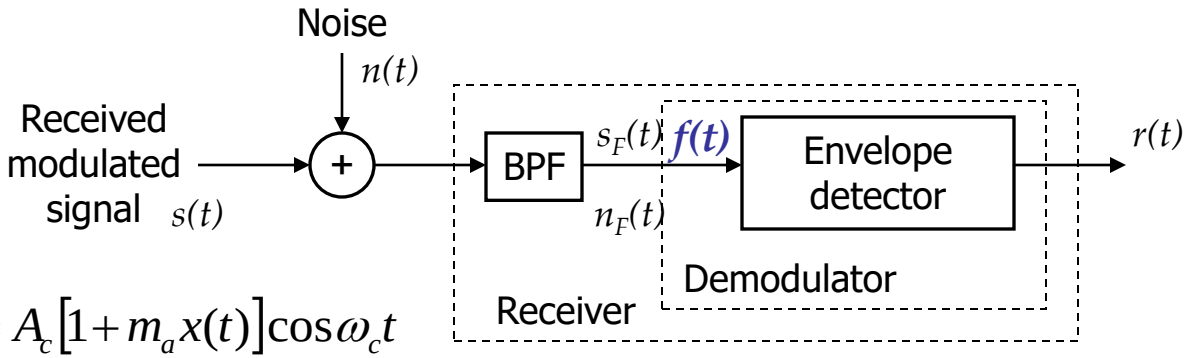
Noise of DSB-LC (AM) Systems

In DSB-LC (AM) system, the received modulated signal is:

$$s(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t$$

where $x(t)$ is the message signal which is band-limited with bandwidth B

Consider the demodulation by ***envelope detection***:



$$s_F(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t$$

$$n_F(t) = n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t$$

$$\overline{n_c^2} = \overline{n_s^2} = \overline{n_F^2} = 2\eta B$$

Input to the envelope detector:

$$f(t) = s_F(t) + n_F(t) = A_c\left[1 + m_a x(t)\right]\cos\omega_c t + \left[n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t\right]$$

Noise of DSB-LC (AM) Systems

Input of the envelope detector:

$$f(t) = A_c\left[1+m_a x(t)\right]\cos\omega_c t + \left[n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t\right]$$
$$= \left\{A_c\left[1+m_a x(t)\right]+n_c(t)\right\}\cos\omega_c t - n_s(t)\sin\omega_c t$$

Output of the envelope detector: $r(t) = |r(t)|e^{j\phi(t)}$

where

$$|r(t)| = \sqrt{\left\{A_c\left[1+m_a x(t)\right]+n_c(t)\right\}^2 + n_s^2(t)} \quad ; \quad \phi(t) = \tan^{-1}\frac{n_s(t)}{A_c\left[1+m_a x(t)\right]+n_c(t)}$$

For $A_c\left[1+m_a x(t)\right] >> \quad n_c(t), \quad n_s(t)$ (for large input SNR):

$$\Rightarrow \quad |r(t)| \approx A_c\left[1+m_a x(t)\right]+n_c(t)$$

Removing the DC component, gives $|r(t)| \approx A_c m_a x(t) + n_c(t)$

which is basically the same as the output signal, $y(t)+n_o(t)$, for synchronous detection of DSB-LC signals without the scaling factor ½.

Thus, ***for high SNR at receiver input, the performance of synchronous detector*** and ***envelope detector is the same***.

Noise in DSB-LC (AM) Systems

For $A_c[1+m_a x(t)] << n_c(t),\ n_s(t)$ (**for small input SNR**):

$$|r(t)| = \sqrt{\{A_c[1+m_a x(t)]+n_c(t)\}^2 + n_s^2(t)}$$

$$= \sqrt{A_c^2[1+m_a x(t)]^2 + n_c^2(t) + n_s^2(t) + 2A_c n_c(t)[1+m_a x(t)]}$$

$$\approx \sqrt{n_c^2(t) + n_s^2(t) + 2A_c n_c(t)[1+m_a x(t)]} \quad \text{for small } A_c[1+m_a x(t)]$$

$$= \sqrt{v_n^2(t)\left\{1+\frac{2A_c n_c(t)}{v_n^2(t)}[1+m_a x(t)]\right\}} \qquad \text{let } v_n^2(t) = n_c^2(t) + n_s^2(t)$$

$$= v_n(t)\sqrt{1+\frac{2A_c n_c(t)}{v_n^2(t)}[1+m_a x(t)]}$$

$$\approx v_n(t)\left\{1+\frac{A_c n_c(t)}{v_n^2(t)}[1+m_a x(t)]\right\} \quad \text{using } \sqrt{1+\alpha} \approx 1+\frac{\alpha}{2} \text{ for small } \alpha$$

$$= v_n(t) + \frac{A_c n_c(t)}{v_n(t)}[1+m_a x(t)]$$

*Thus, for small SNR at receiver input, the **signal and noise are no longer additive**. The signal multiplied by noise cannot be distinguishable. No meaningful output SNR can be defined.*

Noise in Amplitude Modulated Systems

Example: Assuming sinusoidal modulation, show that, in an AM system with envelope detection, the output SNR is given by

$$SNR_o = \frac{m_a^2}{2 + m_a^2} \frac{P_R}{\eta B}$$

where m_a is the modulation index for AM.

Solution: For sinusoidal modulation, $x(t) = \cos \omega_m t$

$$\overline{x^2(t)} = \overline{\cos^2 \omega_m t} = \tfrac{1}{2}$$

Using $SNR_o = \dfrac{m_a^2 \overline{x^2(t)}}{1 + m_a^2 \overline{x^2(t)}} \cdot \dfrac{P_R}{\eta B}$ for AM or DSB-LC

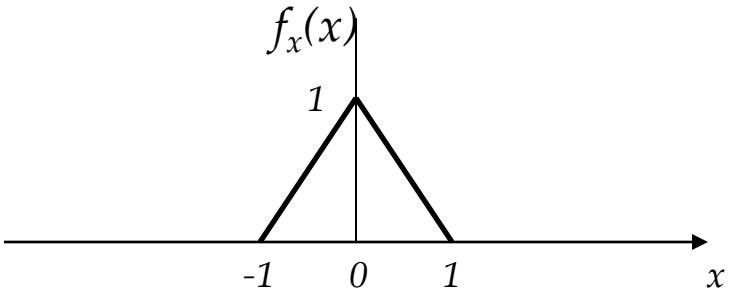
$$SNR_o = \frac{m_a^2 \left(\frac{1}{2}\right)}{1 + \left(\frac{1}{2}\right) m_a^2} \cdot \frac{P_R}{\eta B}$$

Thus,

$$SNR_o = \frac{m_a^2}{2 + m_a^2} \frac{P_R}{\eta B}$$

Noise in Amplitude Modulated Systems

Example: Consider an AM system with additive thermal noise having a power spectral density $\eta/2=10^{-12}$ W/Hz. Assume that the baseband message signal $x(t)$ has a bandwidth of 4kHz and the amplitude distribution shown by the figure. The signal is demodulated by envelope detection and appropriate post-detection filtering. Assume modulation index m_a=1.



Find the minimum value of the carrier amplitude A_c that will yield $SNR_o \geq$ 40dB.

Noise in Amplitude Modulated Systems

Solution:

$$s(t) = A_c[1 + x(t)]\cos\omega_c t$$

$$\overline{x^2(t)} = \int_{-\infty}^{\infty} x^2 f_x(x)dx = 2\int_0^1 x^2(-x+1)dx = \frac{1}{6}$$

For m_a=1, B=4kHz, $\eta/2$=10^{-12} W/Hz ,

$$SNR_o = \frac{m_a^2\overline{x^2(t)}}{1+m_a^2\overline{x^2(t)}}\cdot\frac{P_R}{\eta B} = \frac{(1)(\frac{1}{6})}{1+(1)(\frac{1}{6})}\cdot\frac{P_R}{(2\times10^{-12})(4\times10^3)} = \frac{\frac{1}{7}}{8\times10^{-9}}P_R$$

with $$P_R = \tfrac{1}{2}A_c^2\left[1+m_a^2\overline{x^2(t)}\right] = \tfrac{1}{2}A_c^2\left[1+(1)(\tfrac{1}{6})\right] = \tfrac{7}{12}A_c^2$$

For $$SNR_o \geq 40dB$$

$$SNR_o \geq 10^4$$

$$\frac{\frac{1}{7}}{8\times10^{-9}}\left[\tfrac{7}{12}A_c^2\right] \geq 10^4$$

$$A_c \geq 31\times10^{-3}\ V$$

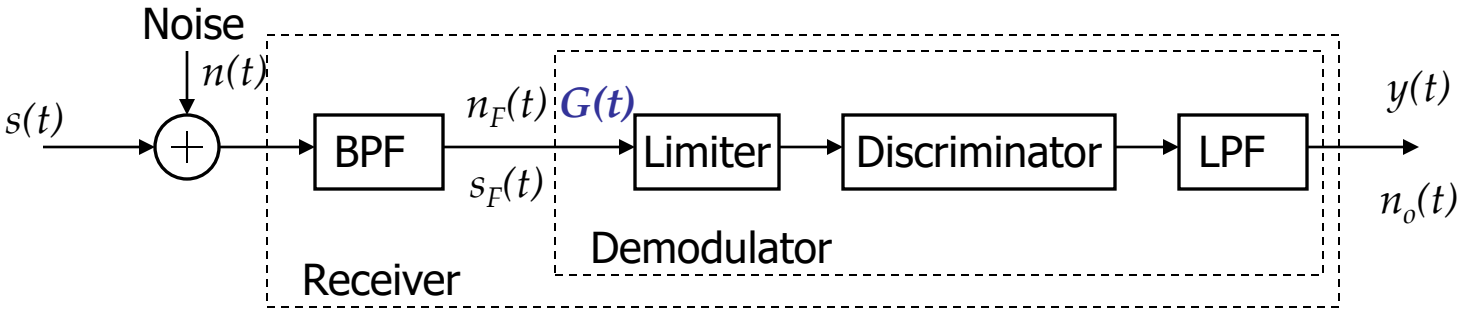
Thus the minimum value of A_c required is 31mV.

Noise in Angle Modulated Systems

For message signal $x(t)$, angle-modulated signal: $s(t) = A_c \cos\left[\omega_c t + \phi(t)\right]$

$$\phi(t) = \begin{cases} k_p x(t) & \text{for PM} \\ k_f \int_0^t x(\tau) d\tau & \text{for FM} \end{cases}$$

The BPF (pre-detection filter) has a bandwidth of $W=2(D+1)B$ where D is the frequency deviation ratio and B is the bandwidth of the message signal.



Input signal power to the demodulator: $\overline{s_F^2(t)} = A_c^2 \overline{\cos^2\left[\omega_c t + \phi(t)\right]} = \frac{1}{2} A_c^2$

Input noise power to the demodulator: $\overline{n_F^2(t)} = \left(\frac{\eta}{2}\right)(2W) = \eta W$

SNR **before** the demodulator: $SNR_i = \dfrac{\overline{s_F^2(t)}}{\overline{n_F^2(t)}} = \dfrac{A_c^2}{2\eta W}$

Noise in Angle Modulated Systems

Let $G(t)$ the input to the demodulator such that

$$G(t) = s_F(t) + n_F(t) = \{A_c \cos[\omega_c t + \phi(t)]\} + \{n_c(t)\cos\omega_c t - n_s(t)\sin\omega_c t\}$$

To facilitate the analysis of the noise only, we can assume $\phi(t)=0$

Thus,
$$\begin{aligned} G(t) &= [A_c + n_c(t)]\cos\omega_c t - n_s(t)\sin\omega_c t \\ &= R(t)\cos[\omega_c t + \theta(t)] \end{aligned}$$

where
$$R(t) = \sqrt{[A_c + n_c(t)]^2 + n_s^2(t)} \;\; ; \;\; \theta(t) = \tan^{-1}\frac{n_s(t)}{A_c + n_c(t)}$$

The limiter suppresses any amplitude variation of $G(t)$, thus in angle modulation, SNR is derived from consideration of $\theta(t)$ only.

$$\theta(t) = \tan^{-1}\frac{n_s(t)}{A_c + n_c(t)}$$

$$\approx \frac{n_s(t)}{A_c + n_c(t)} \qquad \because \tan\sigma \approx \sigma \quad \text{for small } \sigma$$

Noise in Angle Modulated Systems

$$\theta(t) \approx \frac{n_s(t)}{A_c + n_c(t)}$$

For high input SNR, $A_c >> |n_c(t)|$, $A_c >> |n_s(t)|$, thus $\theta(t) \approx \frac{n_s(t)}{A_c}$, $R(t) \gg A_c = a$

Output noise from Limiter: $G(t) = \alpha \cos\left[\omega_c t + \frac{n_s(t)}{A_c}\right]$

Output noise from Discriminator (neglect DC term): $n_d(t) = \frac{\alpha}{A_c}\frac{d}{dt} n_s(t)$

This operation is just like passing $n_s(t)$ through a differentiator with transfer response $H(\omega) = \frac{j\omega\alpha}{A_c}$

Thus, consider the relation of power spectral densities:

$$S_{n_d}(\omega) = |H(\omega)|^2 S_{n_s}(\omega)$$

If the input noise is white with $S_n(\omega) = \eta/2$,

$$S_{n_s}(\omega) = \left[S_n(\omega - \omega_c) + S_n(\omega + \omega_c)\right]_{\text{LP}} = \eta/2 + \eta/2 = \eta$$

Noise in Angle Modulated Systems

With $S_{n_d}(\omega) = |H(\omega)|^2 S_{n_s}(\omega)$ and $H(\omega) = \dfrac{j\omega\alpha}{A_c}$, $S_{n_s}(\omega) = \eta$

Thus, $$S_{n_d}(\omega) = \frac{\alpha^2\omega^2}{A_c^2}\eta \quad \text{for} \quad |f| \le \frac{W}{2}$$

Output noise power after LPF (with bandwidth B):

$$\overline{n_o^2(t)} = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} S_{n_d}(\omega)d\omega = \frac{\alpha^2\eta}{A_c^2}\frac{1}{2\pi}\int_{-2\pi B}^{2\pi B}\omega^2 d\omega = \frac{8\pi^2\alpha^2\eta B^3}{3A_c^2}$$

Output signal from discriminator:

$$y(t) = A_c w_c \left[1 + \frac{k_f}{w_c}x(t)\right] = a w_c \left[1 + \frac{k_f}{w_c}x(t)\right]$$

Neglecting the DC term, gives $$y(t) = \alpha k_f x(t) \Rightarrow \overline{y^2(t)} = \alpha^2 k_f^2 \overline{x^2(t)}$$

Output SNR:

$$SNR_o = \frac{\overline{y^2(t)}}{\overline{n_o^2(t)}} = \frac{\alpha^2 k_f^2 \overline{x^2(t)}}{\left(8\pi^2\alpha^2\eta B^3\right)/\left(3A_c^2\right)} = \frac{3A_c^2 k_f^2 \overline{x^2(t)}}{8\pi^2\eta B^3}$$

Noise in Angle Modulated Systems

$$SNR_o = \frac{3A_c^2 k_f^2 \overline{x^2(t)}}{8\pi^2 \eta B^3}$$

If the message signal is sinusoidal (single-tone), i.e. $x(t) = a_m \cos \omega_m t$

and produces a frequency deviation of $\Delta\omega$,

FM signal: $s(t) = A_c \cos(\omega_c t + \underbrace{\frac{\Delta\omega}{\omega_m} \sin \omega_m t}_{\phi(t)})$

Differentiating $\phi(t)$, gives $k_f x(t) = \Delta\omega \cos \omega_m t$

$\because s(t) = A_c \cos(\omega_c t + \underbrace{k_f \int_0^t x(\tau) d\tau}_{\phi(t)})$

Thus, $k_f^2 \overline{x^2(t)} = (\Delta\omega)^2 \overline{\cos^2 \omega_m t} = \frac{1}{2}(\Delta\omega)^2 = \frac{1}{2}(2\pi\Delta f)^2$

$$\Rightarrow SNR_o = \frac{3}{2}\left(\frac{\Delta f}{B}\right)^2 \frac{A_c^2/2}{\eta B} = \frac{3}{2}\beta^2 \frac{P_R}{\eta B}$$

$$\boxed{SNR_o = \frac{3}{2}\beta^2 \frac{P_R}{\eta B}} \quad \textbf{for FM}$$

Note: When β=5, output SNR of FM system is **37.5 times** that of a baseband system, but the bandwidth of FM system is about 12 times that of the baseband system.

SNR in AM & FM Systems

Under the most favorable conditions in AM, the modulation index is 100%, thus

$$\left(SNR_o\right)_{AM} = \frac{m_a^2}{2+m_a^2}\frac{P_R}{\eta B} = \frac{1}{3}\frac{P_R}{\eta B}$$

For FM:

$$\left(SNR_o\right)_{FM} = \frac{3}{2}\beta^2\frac{P_R}{\eta B}$$

It shows that the output signal-to-noise ratio can be made much higher in FM than in AM by **increasing the modulation index β**.

An increase in β also increases the bandwidth so that FM systems provide an improvement in signal-to-noise at the expense of an increase in bandwidth.

To realize any signal-to-noise improvement in FM over AM we must have

$$\frac{3}{2}\beta^2 > \frac{1}{3} \quad \Rightarrow \quad \beta > 0.47$$

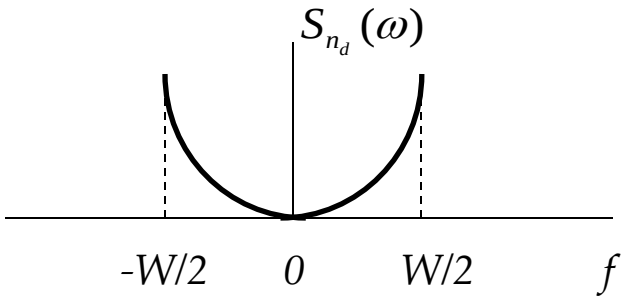
This condition is approximately the transition point between narrowband and wideband FM.

Narrowband FM provides no signal-to-noise improvement over AM.

Pre-emphasis and De-emphasis in FM

Recall: the **power spectral density at the FM discriminator output**

$$S_{n_d}(\omega) = \frac{\alpha^2 \omega^2}{A_c^2}\eta \quad \text{for} \quad |f| \le \frac{W}{2}$$

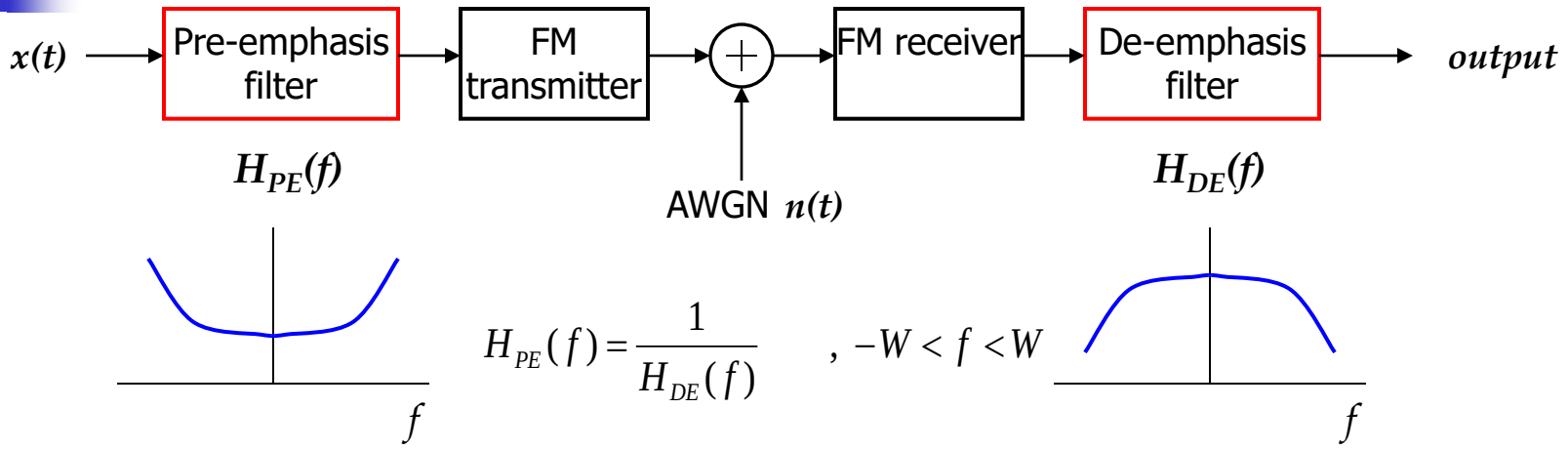


Noise power is larger at higher frequencies within the message bandwidth.

Pre-emphasis: the high-frequency components in the input message signal are emphasized at the transmitter before the noise is introduced.

De-emphasis: at the output of FM demodulator, the inverse operation is performed to de-emphasize the high frequency components

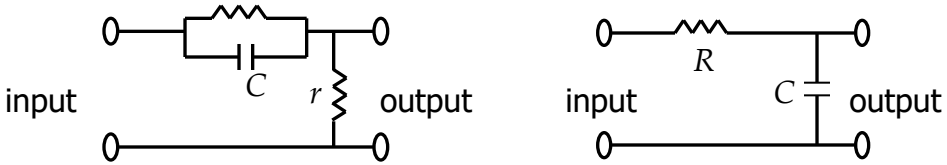
Pre-emphasis and De-emphasis in FM



$$H_{PE}(f)=\frac{1}{H_{DE}(f)} \quad , -W<f<W$$

Example:

$$H_{PE}(f)=K\left(1+\frac{jf}{f_o}\right) \quad ; \quad H_{DE}(f)=\frac{1/K}{1+jf/f_o}$$



Output noise power without de-emphasis filter:

$$\overline{n_o^2(t)}=\frac{1}{2\pi}\int_{-2\pi B}^{2\pi B}S_{n_d}(\omega)d\omega=\frac{\alpha^2\eta}{A_c^2}\frac{1}{2\pi}\int_{-2\pi B}^{2\pi B}\omega^2 d\omega$$

Output noise power with de-emphasis filter:

$$\overline{n_o^2(t)}=\frac{1}{2\pi}\int_{-2\pi B}^{2\pi B}\left|\frac{1}{H_{PE}(\omega)}\right|^2 S_{n_d}(\omega)d\omega=\frac{\alpha^2\eta}{A_c^2}\frac{1}{2\pi}\int_{-2\pi B}^{2\pi B}\omega^2\left|\frac{1}{H_{PE}(\omega)}\right|^2 d\omega$$